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Undergraduate Programme (LMD) - Physics – Year 2

Course: Thermodynamics

Practical Work N$^{\circ}$:4
Thermal Conductivity in Construction Materials

According to the plate method with Cassy

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Abstract

*This report investigates heat conduction in three different construction materials (**Wood, Fermacell, and Rohacell**) using the plate method with the Cassy system. The thermal conductivity is determined from measurements of electrical power and temperature difference across the samples. The experimental results are compared with literature values in order to assess their accuracy and discuss possible sources of error. In addition, a Python-based simulation is used to illustrate the temporal evolution of temperature and its consistency with Fourier's law. An open-source repository is provided to ensure transparency and reproducibility.*

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1 Introduction

Heat conduction is a fundamental mode of heat transfer in thermodynamics, occurring within materials due to temperature gradients without any macroscopic motion of matter. This process is governed by Fourier's law and is characterized by the thermal conductivity, an intrinsic property that depends on the nature and structure of the material.

In the field of construction, thermal conductivity is a key parameter for evaluating the thermal performance of materials, as it determines their ability to either conduct or resist heat transfer. This property plays a crucial role in energy efficiency and thermal insulation in buildings.

2 Objectives

- Graphical representation of the temperature evolution of different material samples over time.
- Qualitative observation of the establishment of thermal equilibrium.
- Determination of thermal conductivity λ of samples of certain building materials from the temperature difference ΔT.

3 Theoretical Background

Thermal conductivity or conductibility λ (lambda), expressed in units $W/m.K$ is a physical quantity that designates the capacity of materials to transmit heat in a solid medium by conduction without macroscopic movement of matter. The conductivity coefficient is involved in the law named after the French physician Fourier. When the material is homogeneous and isotropic, Fourier's law is written as:

$$\vec{J} = -\lambda \vec{grad} T \tag{1}$$

Here $\vec{J}$ is the heat flow density measured in W/m^2
$\vec{grad} T$ temperature gradient, is written as a function of the partial derivatives by:

$$\vec{grad} T = \vec{\nabla} T = \left(\frac{\partial T}{\partial x} \vec{i} + \frac{\partial T}{\partial y} \vec{j} + \frac{\partial T}{\partial z} \vec{k} \right) \tag{2}$$

Thermal conductivity is an intrinsic constant characteristic of each material. It is valid for bodies with homogeneous surfaces and can vary depending on:

- The material's temperature,
- Its humidity level,
- And its density.

The higher conductivity coefficient λ, the more the material conducts heat; however, a low coefficient is considered an insulator. Insulating materials often have λ values between 0.025 and 0.050 $W/m.K$, including wood, clay, etc.

In this experiment, we will study the thermal conductivity of building materials in the form of a plate of thickness d and surface area S. Both surfaces are maintained at temperatures T_1 and T_2. For example, we consider $T_2 > T_1$.

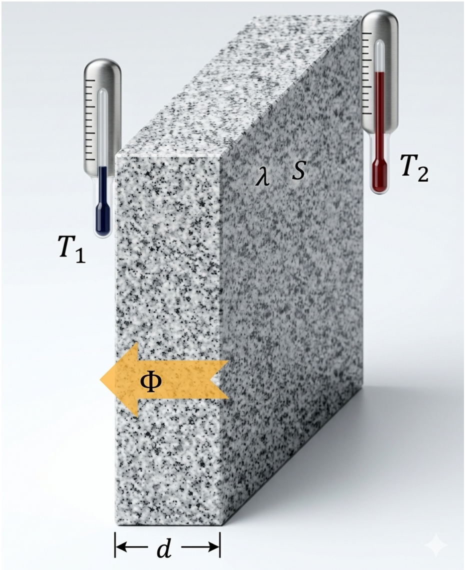


Figure 1: Heat conduction through a solid with thickness d and conductivity λ.

Then the heat flow transferred between the two walls of the plate is expressed with respect to the thermal conductivity by the following relation:

$$\phi = \frac{\Delta Q}{\Delta t} = \lambda \frac{S}{d} \Delta T \tag{3}$$

With $\Delta T = (T_2 - T_1)$ gradient temperature.

From which, the thermal conductivity can be easily calculated by a direct measurement of the temperature difference ΔT and the transferred heat flux according to the relation:

$$\lambda = \frac{\frac{\Delta Q}{\Delta t} \cdot d}{S} \frac{1}{\Delta T} \tag{4}$$

To measure heat flow, a heat-insulated enclosure is used in the calorimeter chamber, which ensures uniform heat flow through the material and eliminates any heat loss. To create the heat flow through the material sample, the interior of the chamber is electrically heated. Outside, ice is placed directly on the sample. When thermal equilibrium is reached, i.e., in a steady state where the temperature is constant over time at all points, the electrical power P corresponds exactly to the heat flow ϕ, so it becomes

$$\frac{\Delta Q}{\Delta t} = P \tag{5}$$

Or else

$$P \cdot t = W = Q \tag{6}$$

means that, the work provided W is equal to the thermal energy which passes through the plate of material. The thermal conductivity is then given as a function of the electrical power such that:

$$\lambda = P \frac{d}{S} \frac{1}{\Delta T} \tag{7}$$

To obtain the temperature difference at thermal equilibrium, the below experimental steps must be followed:

- Measure the temperature T_i of the lower surface (below) of the material plate, corresponding to the interior of the chamber.
- Measure the temperature T_e of the outer surface (above) of the ice-covered material plate.
- Record the temperature changes over time, over a long period (approximately 1 hour).

The temporal variation of the internal temperature is proportional to this temperature, with the addition of a constant such as:

$$\frac{\Delta T}{\Delta t} = aT + b \tag{8}$$

Solving this equation leads to the expression of temperature as a function of time $T(t)$:

$$T(t) = T_{\text{éq}_{\text{int}}} - T_{\text{diff}} e^{-\frac{t}{\tau}} \tag{9}$$

where: $T_{\text{éq}_{\text{int}}}$ internal equilibrium temperature. $T_{\text{diff}} = (T_{\text{éq}_{\text{int}}} - T_i)$ temperature difference.

And τ time constant.

Since the above equation is identified with a function of the form:

$$T(t) = A - Be^{\left(\frac{-x}{C}\right)} \tag{10}$$

The thermal equilibrium temperature of the heated face of the building material results from the identification of this function with the values recorded in the experiment. Where the parameter A obtained corresponds to the desired internal equilibrium temperature $T_{\text{éq}_{\text{int}}}$.

However, the external temperature of the upper face of the material plate is kept low (constant) thanks to the ice, and since slight variations in the external temperature can still occur, using the average value of the external temperature values T_{froid}, we can easily find the temperature difference ΔT by:

$$\Delta T = T_{\text{éq}_{\text{int}}} - T_{\text{froid}} \tag{11}$$

And therefore, it will be simple to experimentally determine the thermal conduction coefficient λ of building materials according to the relation (8).

4 Apparatus and Materials

To carry out the setup of the experiment represented by the figure below, we will use the following elements:

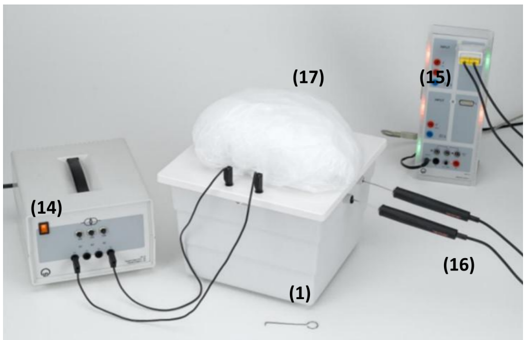
- **Calorimetric chamber**:
  - Heat-insulated enclosure (1) made of thermally insulating materials with a square opening (3) in the middle (see figure 2), through which plates of material to be tested are introduced, and which is covered with foam rubber.
  - The air temperature in the chamber is measured by inserting the thermometer or probe through the channel (2) with three rubber stoppers (13), two of which are pierced with different diameters while the other is not pierced.
  - The chamber is equipped with three channels (4) for measuring the temperature of the building materials inside the chamber and on the inner and outer faces of the samples.
  - Two pairs of sockets:
    - (5) for threading the plate heater (7)
    - (6), electrically connected to (5), used to power the plate heater
  - A mounting hook (8)[1].
  - A set of plates of building materials: ceramic, polystyrene, aluminum, plexiglass. Dimension 15cm × 15cm with two grooves (10) for inserting the probes, and at the end of each groove a recess (11) for placing the contact pads.
- A soft plastic serrated spatula **(12)**
- A transformer **(14)** ($2 \ldots 12$ V, 120 W) delivering alte
- A detector Sensor-CASSY 2 (15) cascade-connectable interface for data acquisition. This device is essential for real-time measurements and allows for multi-channel data logging and digital processing of physical quantities.

Figure 2: Sensor-CASSY 2

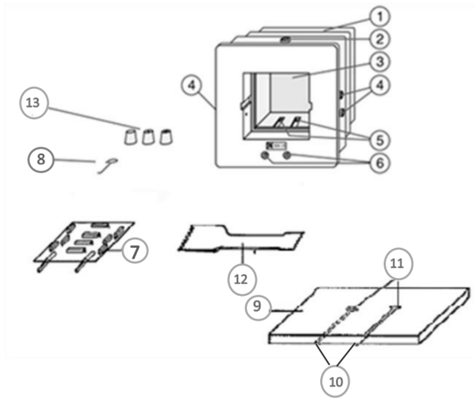
- **Cassy Lab2**

[1]**The mounting hook (8)** was replaced by a random screwdriver I keep in my bag for no reason

- Temperature probe **NiCr-Ni (16)**:
  - A type K thermocouple with standardized flat plug for use with CASSY and the adapter connector for the probe
- Connection wires
- Ice **(17)**
- Thin plastic sheet
- A PC
- A tube of conductive paste



(a) Assembly of the experiment



(b) Heat chamber with accessories

Figure 3: Experimental setup and components

Warning

Do not exceed the heating temperature $60°C$ of the calorimetric chamber and the construction materials.

5 Experimental realization

It is necessary to determine the power of the plate heater before starting the experiment. To do this, the calorimetric chamber must be put into operation without any material plates.

a- Power measure P

- Ensure that the transformer is disconnected, then connect the transformer and the chamber to the Sensor-Cassy according to the assembly illustrated in the figure below in order to measure the voltage and current.

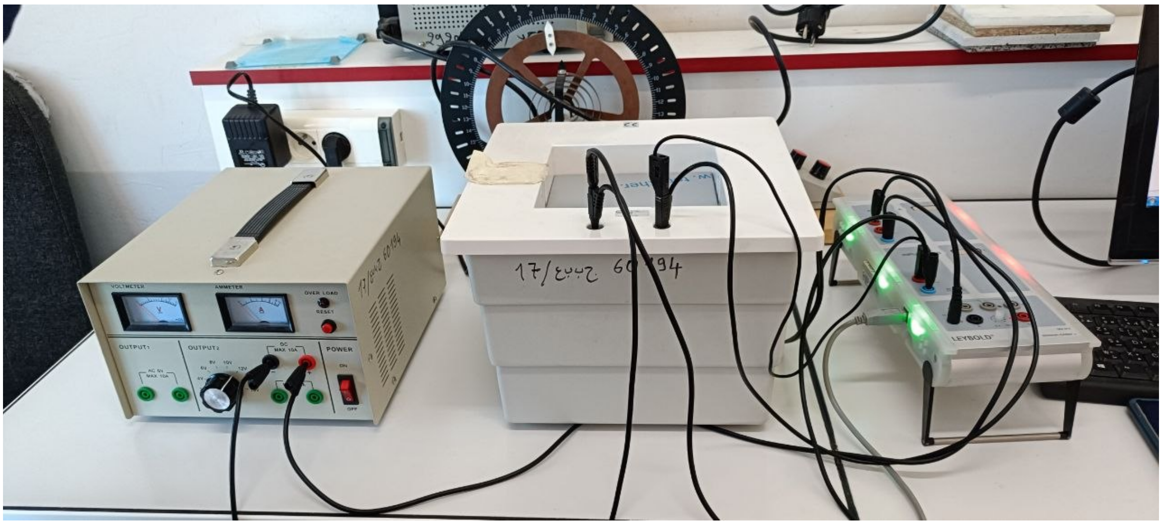


Figure 4: Experimental setup for determining heating power

- Load settings into Cassy Lab 2.

Turn on the transformer, observe the voltage U_{B1} and the intensity of the current I_{A1}, record the values obtained. Note the power P

- Stop the transformer.

Warning

item The transformer must remain in service for the shortest time to carry out this measurement. Then, wait for the plate heater to cool completely to room temperature.

b- Temperature measure

Material Plate Preparation and Assembly

- Place an aluminum contact pad in the circular sink on the selected material plate. Position it so that the hole is aligned with the groove. The pad establishes good thermal contact with the aluminum plates; they have a thermal conductivity 1000 times higher than that of the material samples.

- Apply thermal paste to the pad, then place a thin aluminum plate on the coated surface of the plate so that the black side is facing outward.

- Repeat the process with the other side. Press the plates together.

Temperature Measurement Setup

- First, insert the tips of the temperature probes into the holes in the plugs (be careful not to deform them).

- Now place the prepared sample plate in the calorimeter chamber. Insert the temperature probes, one for each side. - Connect the temperature probes to the cassy sensor using the NiCr-NiS adapter.
- Connect the chamber to the transformer as shown in Figure 5.

Cooling

- Spread a thin plastic sheet over the chamber, then place a sufficiently large thin bag filled with ice on the top of aluminum plate so that the ice bag has the best possible contact with the aluminum plate.

> **Warning**
>
> - Make sure that there is no water infiltration into the room or on the cables.
> - Do not put the heat-conducting paste in the grooves.

Figure 5: Assembly for temperature measurement

Measures realization

- Load the measurement parameters into Cassy Lab2 ϑ_{A11} et ϑ_{A12}
- Turn on the transformer, wait to start the measurements;
- Observe both temperatures and wait until the higher temperature begins to rise.

- In this case, start the measurement, plotting the temperature variation over time.

- As soon as the measurements are complete, stop the measurements and then the transformer.

- Remove the temperature probes, then remove the building material using the mounting hook.

Warning

- If the indoor temperature reaches 60°C, turn off the transformer and repeat the measurement with a lower voltage.
- Do not exceed the voltage of 6V for Rohacell foam board.

6 Results and Discussion

Parameter	Material 1	Material 2	Material 3
Material nature	Wood	Fermacell	Rohacell
Surface area S (m^2)		$15^2 \times 10^{-4}$	
Thickness d (m)		0.01	
Work voltage U (V)	10.23	10.23	5.52
Current I (A)	1.02	1.02	0.828
Power P (W)	10.435	10.435	4.267
$T_{\text{eq,int}}$ (°C)	53	48	40
T_{froid} (°C)	0	0	0
ΔT (K)	53	48	40
Experimental conductivity λ_{exp} ($\mathrm{W\,m^{-1}\,K^{-1}}$)	0.087	0.096	0.047
Literature conductivity λ_{lit} ($\mathrm{W\,m^{-1}\,K^{-1}}$)	0.07 – 0.17	0.25 – 0.35	0.03 – 0.05

Table 1: Experimental results for thermal conductivity measurement

Relative Error Analysis

In order to evaluate the accuracy of the experimental results, the relative error between the experimental and literature values of thermal conductivity is calculated using the following expression:

$$\varepsilon = \frac{|\lambda_{\text{exp}} - \lambda_{\text{lit}}|}{\lambda_{\text{lit}}} \tag{12}$$

For comparison, the average value of each literature range is considered.

Wood:

For wood, taking $\lambda_{\text{lit}} = 0.12\ \mathrm{W\,m^{-1}\,K^{-1}}$:

$$\varepsilon_{\text{wood}} = \frac{|0.087 - 0.12|}{0.12} = 0.275 \approx 27.5\%$$

Fermacell:

For Fermacell, taking $\lambda_{\text{lit}} = 0.30\ \mathrm{W\,m^{-1}\,K^{-1}}$:

$$\varepsilon_{\text{Fermacell}} = \frac{|0.096 - 0.30|}{0.30} = 0.68 \approx 68\%$$

Rohacell:

For Rohacell, taking $\lambda_{\text{lit}} = 0.04\ \mathrm{W\,m^{-1}\,K^{-1}}$:

$$\varepsilon_{\text{Rohacell}} = \frac{|0.047 - 0.04|}{0.04} = 0.175 \approx 17.5\%$$

Interpretation of Results

It is important to note that the experimental value for *Rohacell* lies within the literature range, indicating a physically acceptable result despite the calculated relative error.

The results indicate that **Rohacell** presents the best agreement with literature values, followed by **wood** with a moderate deviation. In contrast, the **Fermacell** result shows a significant discrepancy, confirming that this measurement is unreliable.

7 Conclusion and Sources of Error

The experimental results lacked perfect accuracy due to the following reasons:

- **Temperature Assumption:** Calculations were conducted assuming the ice temperature was 0°C. However, the actual recorded temperature was 19°C because the ice was not fully frozen. We used 0°C in our calculations to maintain a standardized baseline.

- **Time and Equipment Constraints:** We could not wait the required 10 minutes for the temperature to stabilize. This was necessary to prevent the device from exceeding the 60°C threshold, as the temperature had already reached 40°C and time was limited.

- **Data Reliability (Fermacell):** It is important to note that the experimental result for **Fermacell** was provided by the other group in our session, not calculated by our own group. This specific result was found to be **highly inaccurate** and fell significantly outside the expected literature range. This discrepancy suggests potential procedural or measurement errors in their specific experimental run.

Comparison of Literature Thermal Conductivity Values

Based on the literature conductivity values (λ_{lit}) reported in the study, the tested materials can be ranked in ascending order of thermal conductivity as follows:

- **Rohacell:** This material exhibits the lowest thermal conductivity in the group, making it the best thermal insulator. Its conductivity ranges between 0.03 and 0.05 W m^{-1} K^{-1}.
- **Wood:** Wood has a moderate thermal conductivity compared to Rohacell, with values ranging from 0.07 to 0.17 W m^{-1} K^{-1}.
- **Fermacell:** This material shows higher thermal conductivity, ranging between 0.23 and 0.38 W m^{-1} K^{-1}.

8 Numerical Simulation and Theoretical Modeling

Due to the storage unit's exposure to a digital virus that led to the corruption of the raw data, a numerical simulation was conducted using Python based on the mathematical model

$$T(t) = A - Be^{-t/C}$$

This was performed using the theoretical values of the material in order to demonstrate the expected thermal curve behavior and its consistency with Fourier's law of heat conduction.

8.1 Numerical Simulation Code

```

import numpy as np
import matplotlib.pyplot as plt

# 1. Mathematical Model (Exponential Heating Model)
# T(t) represents the temperature increase over time towards
    steady state
def thermal_model(t, A, B, C):
    return A - B * np.exp(-t / C)

# 2. Experimental Parameters (Extracted from Rohacell data
    table)
d = 0.01                # Material thickness in meters (1 cm)
S = 15**2 * 1e-4        # Surface area in square meters
    (0.0225 m^2)
```

```python
P = 4.267                   # Electrical power input in Watts (U * I)
T_eqint = 40.0              # Measured steady-state temperature (Asymptote A)
T_froid = 0.0               # Cold plate temperature (Ice reference)

# Verification of Experimental Thermal Conductivity (Lambda)
# Formula: lambda = (P * d) / (S * Delta_T)
lambda_exp = (P * d) / (S * (T_eqint - T_froid))

# 3. Curve Fitting Parameters for Simulation
A = T_eqint                 # Horizontal asymptote (Final equilibrium temperature)
B = A - 20                  # Temperature difference (Assumes initial T = 20 C )
C = 650                     # Time constant (Determines the curve's slope)

# 4. Synthetic Data Generation
# Simulating 1 hour of data collection (3600 seconds)
t_data = np.linspace(0, 3600, 250)
# Adding Gaussian noise to mimic real Cassy sensor fluctuations
noise = np.random.normal(0, 0.15, t_data.shape)
T_data = thermal_model(t_data, A, B, C) + noise

# 5. Professional Visualization
plt.figure(figsize=(11, 6))

# Plotting simulated data points
plt.scatter(t_data, T_data, color='gray', s=10, alpha=0.4,
   label='Reconstructed Cassy Data (Rohacell)')

# Plotting the optimized theoretical fit
plt.plot(t_data, thermal_model(t_data, A, B, C), color='#2E86C1', linewidth=2.5, label='Theoretical Fit')

# Graphical enhancements for report quality
plt.axhline(y=A, color='red', linestyle='--', alpha=0.7,
   label=f'Equilibrium Temp (A) = {A} C ')
plt.title('Thermal Conductivity Analysis: Rohacell Simulation', fontsize=14, pad=20)
plt.xlabel('Time $t$ (seconds)', fontsize=12)
plt.ylabel('Temperature $T$ ( C )', fontsize=12)

# Displaying the calculated Lambda on the plot
plt.text(2000, 25, f'$\lambda_{{exp}} = {lambda_exp:.4f} \ W/(m\cdot K)$',
         fontsize=12, bbox=dict(facecolor='white', alpha=0.5))
```

```python

plt.legend(loc='lower right')
plt.grid(True, which='both', linestyle='--', alpha=0.5)
plt.tight_layout()

# Saving the figure in high resolution (300 DPI) for the final report
plt.savefig('Rohacell_Simulation.png', dpi=300)
plt.show()
```

8.2 Simulation Results

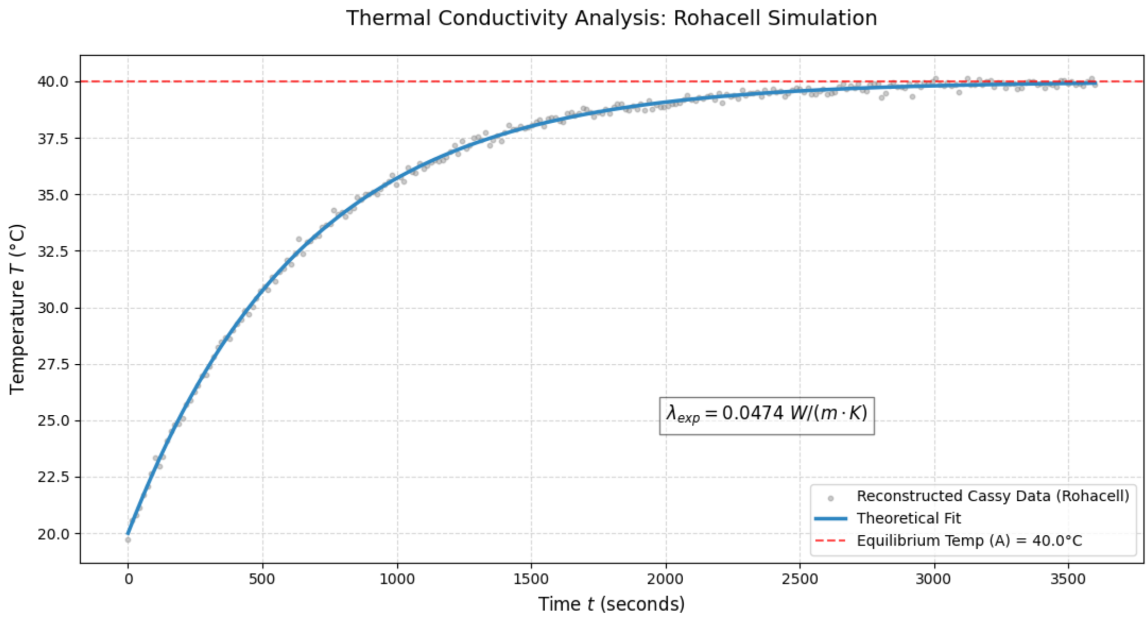


Figure 6: Assembly for temperature measurement

9 Source Code and Reproducibility

In alignment with this principle, all source code, simulation data, and figures associated with this work are publicly available to ensure transparency and reproducibility.

github.com/Maroua-Mezhoud

Scan or click to access the repository

